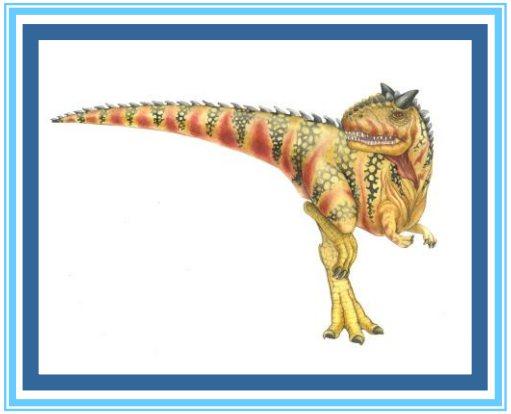
Chapter 10: Virtual Memory



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Chapter 10: Virtual Memory

Background

Demand Paging

Copy-on-Write

Page Replacement

Allocation of Frames

Thrashing

Memory-Mapped Files

Allocating Kernel Memory

Other Considerations

Operating-System Examples

Objectives

To describe the benefits of a virtual memory system

To explain the concepts of demand paging, page-replacement algorithms, and allocation of page frames

To discuss the principle of the working-set model

To examine the relationship between shared memory and memory-mapped files

To explore how kernel memory is managed

Background

- Code needs to be in memory to execute, but entire program rarely used
  - Error code, unusual routines, large data structures
- Entire program code not needed at same time
- Consider ability to execute partially-loaded program
  - Program no longer constrained by limits of physical memory
  - Each program takes less memory while running -> more programs run at the same time
    - Increased CPU utilization and throughput with no increase in response time or turnaround time
  - Less I/O needed to load or swap programs into memory -> each user program runs faster

Background (Cont.)

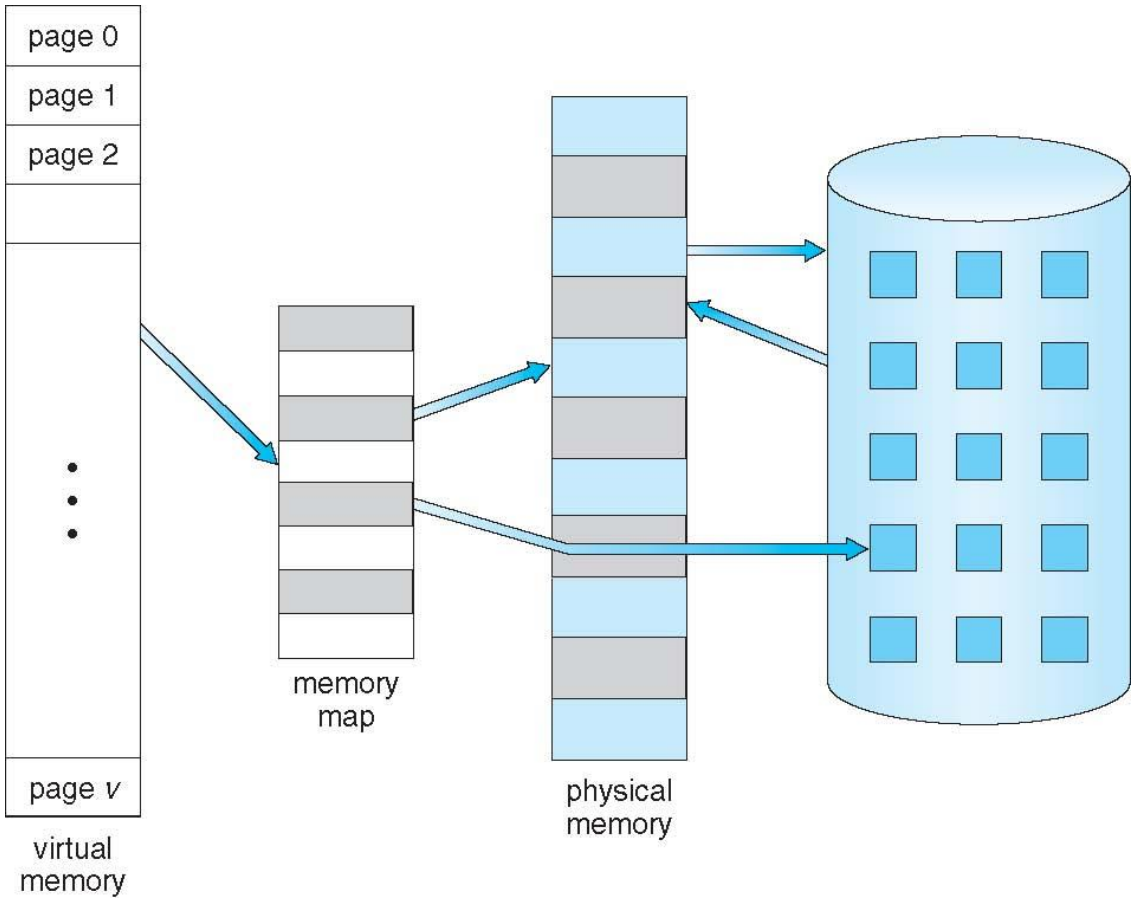
Virtual memory – separation of user logical memory from physical memory

- Only part of the program needs to be in memory for execution
- Logical address space can therefore be much larger than physical address space
- Allows address spaces to be shared by several processes
- Allows for more efficient process creation
- More programs running concurrently
- Less I/O needed to load or swap processes

Background (Cont.)

- **Virtual address space** – logical view of how process is stored in memory
  - Usually start at address 0, contiguous addresses until end of space
  - Meanwhile, physical memory organized in page frames
  - MMU must map logical to physical
- Virtual memory can be implemented via:
  - Demand paging
  - Demand segmentation

Virtual Memory That is Larger Than Physical Memory



Virtual-address Space

Usually design logical address space for stack to start at Max logical address and grow “down” while heap grows “up”

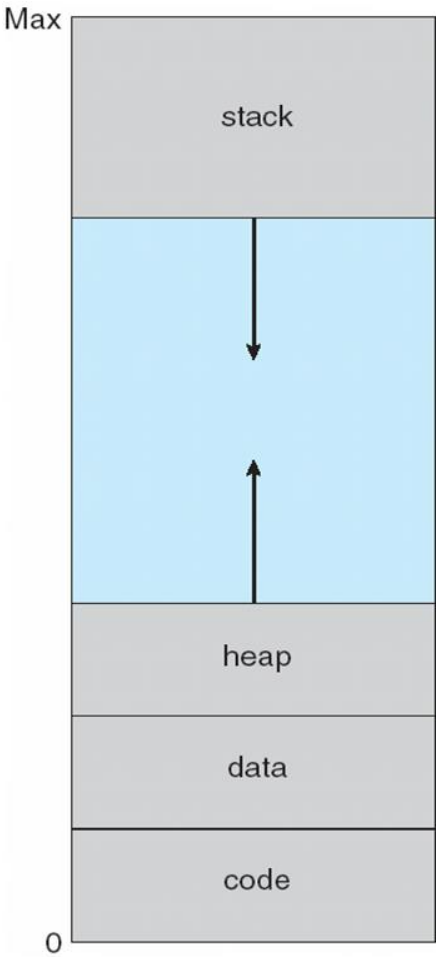
- Maximizes address space use
- Unused address space between the two is hole
  - No physical memory needed until heap or stack grows to a given new page

Enables **sparse** address spaces with holes left for growth, dynamically linked libraries, etc

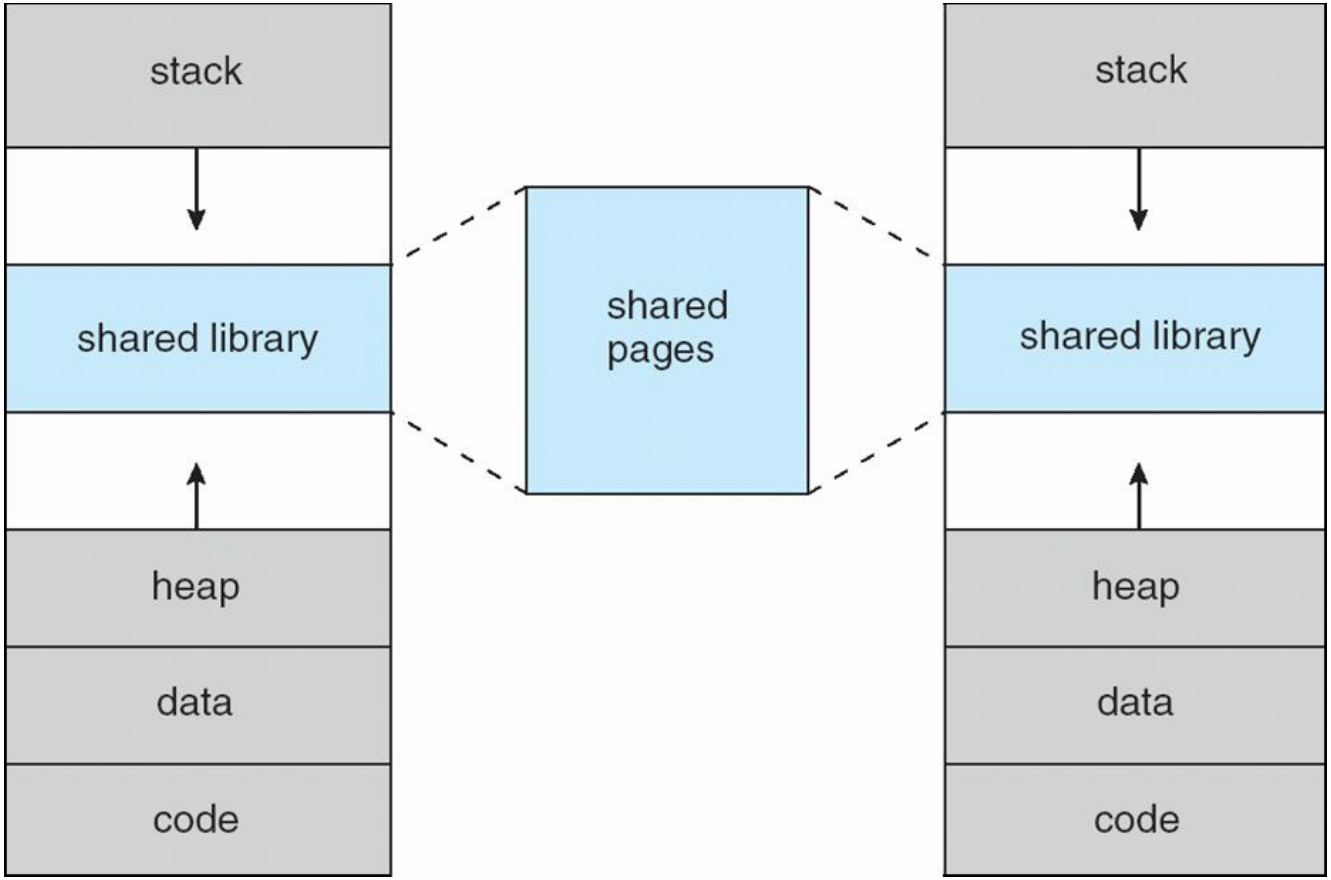
System libraries shared via mapping into virtual address space

Shared memory by mapping pages read-write into virtual address space

Pages can be shared during `fork()`, speeding process creation

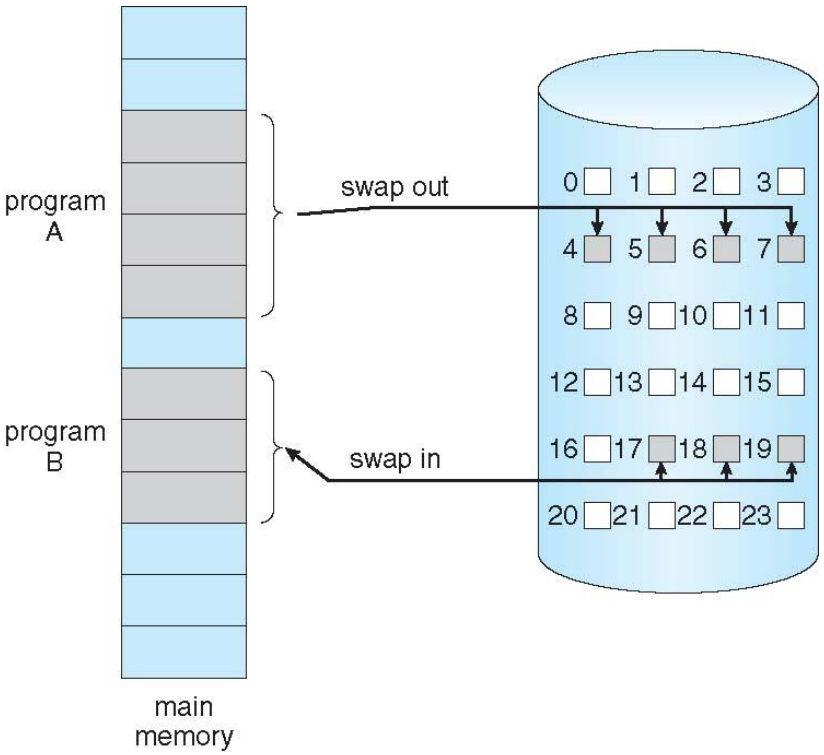


Shared Library Using Virtual Memory



Demand Paging

- Could bring entire process into memory at load time
- Or bring a page into memory only when it is needed
  - Less I/O needed, no unnecessary I/O
  - Less memory needed
  - Faster response
  - More users
- Similar to paging system with swapping (diagram on right)
- Page is needed ⇒ reference to it
  - invalid reference ⇒ abort
  - not-in-memory ⇒ bring to memory
- **Lazy swapper** – never swaps a page into memory unless page will be needed
  - Swapper that deals with pages is a **pager**



Basic Concepts

With swapping, pager guesses which pages will be used before swapping out again

Instead, pager brings in only those pages into memory

How to determine that set of pages?

Need new MMU functionality to implement demand paging

If pages needed are already **memory resident**

No difference from non demand-paging

If page needed and not memory resident

Need to detect and load the page into memory from storage

- Without changing program behavior
- Without programmer needing to change code

Valid-Invalid Bit

With each page table entry a valid–invalid bit is associated (**v** ⇒ in-memory – **memory resident**, **i** ⇒ not-in-memory)

Initially valid–invalid bit is set to **i** on all entries

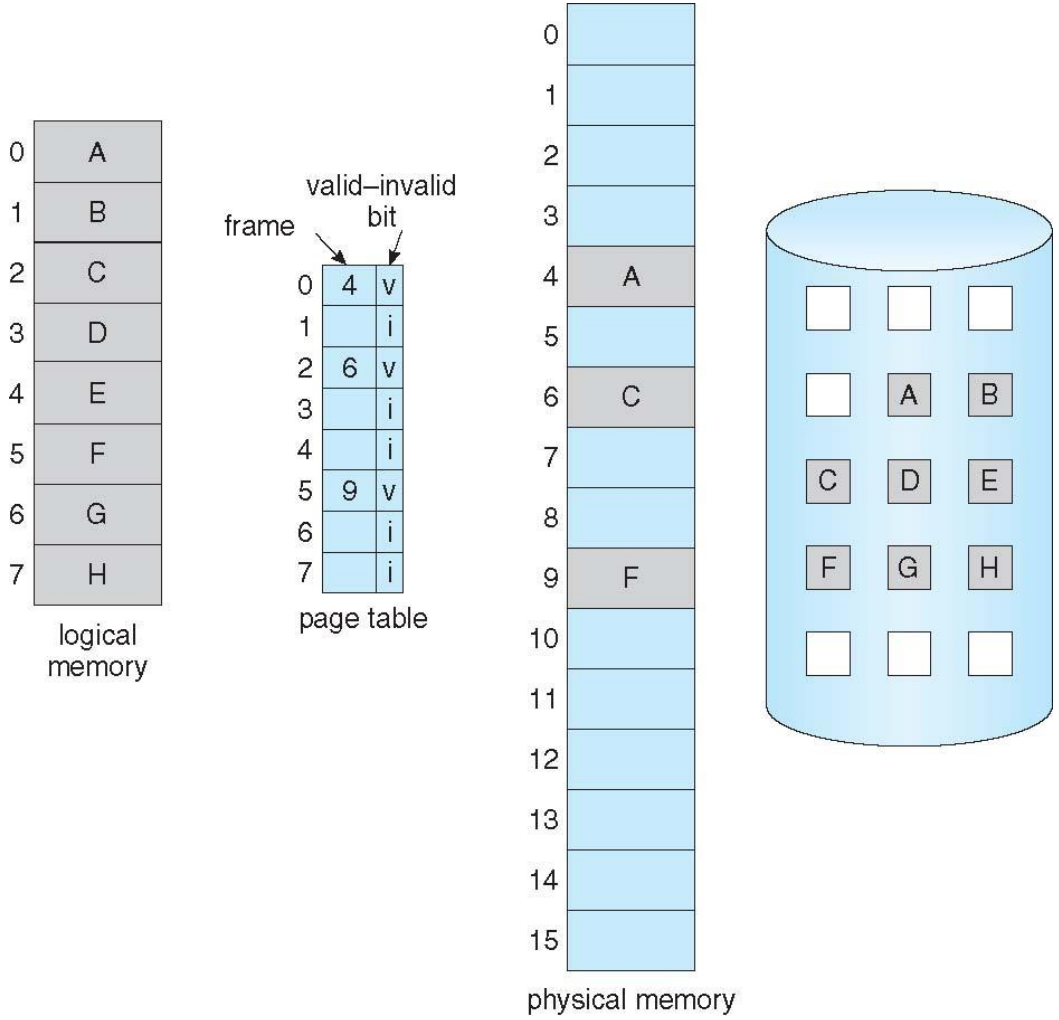
Example of a page table snapshot:

Frame #	valid-invalid bit
	v
	v
	v
	i
. . .	
	i
	i

page table

During MMU address translation, if valid–invalid bit in page table entry is **i** ⇒ page fault

Page Table When Some Pages Are Not in Main Memory



Page Fault

If there is a reference to a page, first reference to that page will trap to operating system:

page fault

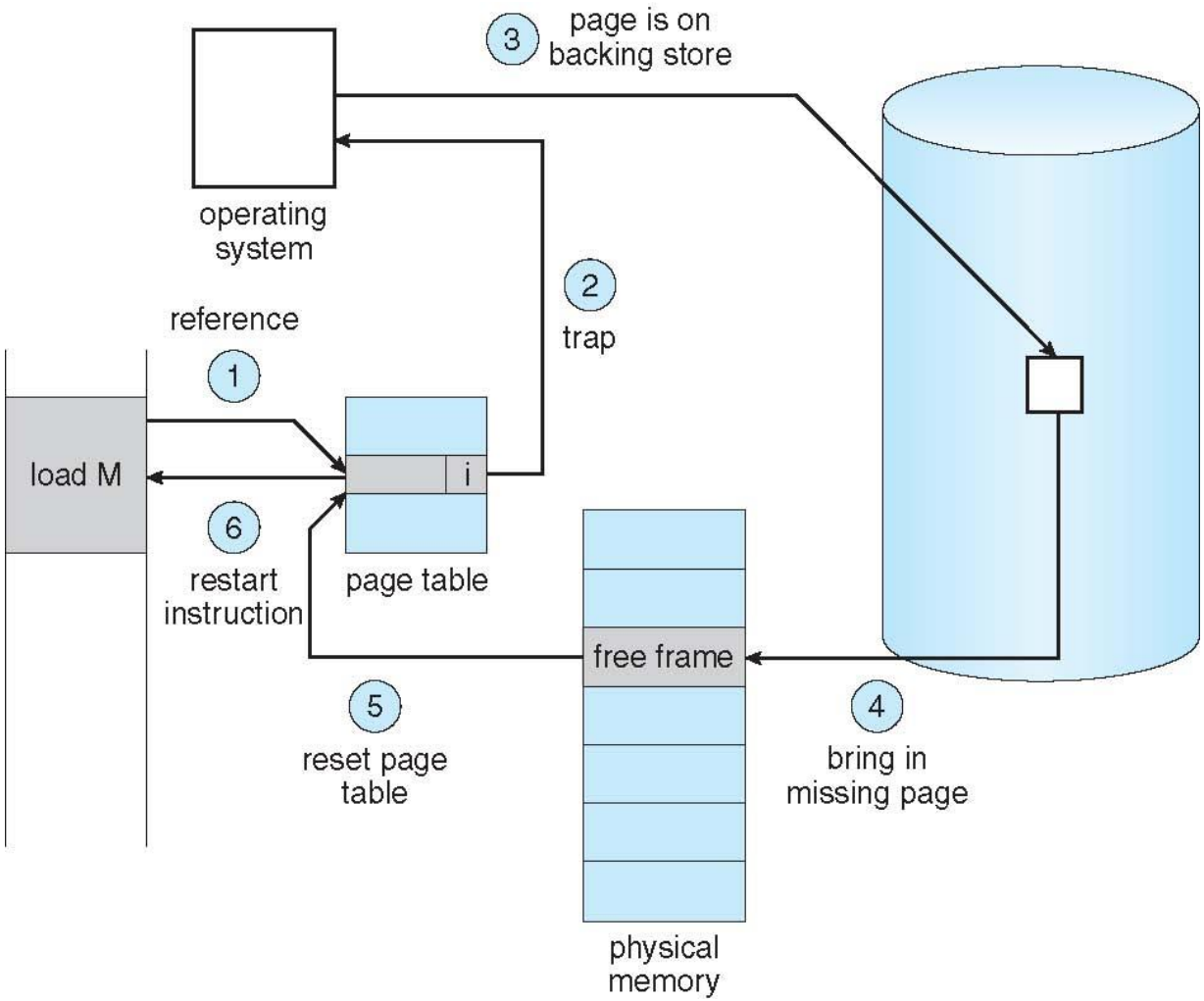
1. Operating system looks at another table to decide:

   Invalid reference $\Rightarrow$ abort

   Just not in memory

2. Find free frame
3. Swap page into frame via scheduled disk operation
4. Reset tables to indicate page now in memory
   Set validation bit = **v**
5. Restart the instruction that caused the page fault

Steps in Handling a Page Fault



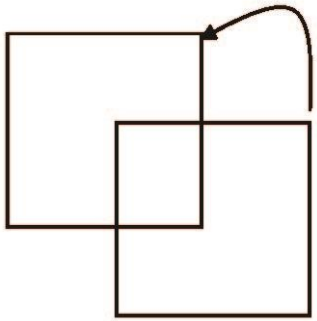
Aspects of Demand Paging

- Extreme case – start process with *no* pages in memory
  - OS sets instruction pointer to first instruction of process, non-memory-resident -> page fault
  - And for every other process pages on first access
  - **Pure demand paging**
- Actually, a given instruction could access multiple pages -> multiple page faults
  - Consider fetch and decode of instruction which adds 2 numbers from memory and stores result back to memory
  - Pain decreased because of **locality of reference**
- Hardware support needed for demand paging
  - Page table with valid / invalid bit
  - Secondary memory (swap device with **swap space**)
  - Instruction restart

Instruction Restart

Consider an instruction that could access several different locations

block move



auto increment/decrement location

Restart the whole operation?

- What if source and destination overlap?

Performance of Demand Paging

Stages in Demand Paging (worse case)

1. Trap to the operating system
2. Save the user registers and process state
3. Determine that the interrupt was a page fault
4. Check that the page reference was legal and determine the location of the page on the disk
5. Issue a read from the disk to a free frame:
    1. Wait in a queue for this device until the read request is serviced
    2. Wait for the device seek and/or latency time
    3. Begin the transfer of the page to a free frame
6. While waiting, allocate the CPU to some other user
7. Receive an interrupt from the disk I/O subsystem (I/O completed)
8. Save the registers and process state for the other user
9. Determine that the interrupt was from the disk
10. Correct the page table and other tables to show page is now in memory
11. Wait for the CPU to be allocated to this process again
12. Restore the user registers, process state, and new page table, and then resume the interrupted instruction

Performance of Demand Paging (Cont.)

Three major activities

- Service the interrupt – careful coding means just several hundred instructions needed
- Read the page – lots of time
- Restart the process – again just a small amount of time

Page Fault Rate $0 \le p \le 1$

- if $p = 0$ no page faults
- if $p = 1$, every reference is a fault

Effective Access Time (EAT)

$$\begin{aligned} EAT = &(1 - p) \times \text{memory access} \\ &+ p\,(\text{page fault overhead} \\ &\quad + \text{swap page out} \\ &\quad + \text{swap page in}) \end{aligned}$$

Demand Paging Example

Memory access time = 200 nanoseconds

Average page-fault service time = 8 milliseconds

EAT = (1 – p) x 200 + p (8 milliseconds)

= (1 – p x 200 + p x 8,000,000

= 200 + p x 7,999,800

If one access out of 1,000 causes a page fault, then

EAT = 8.2 microseconds.

This is a slowdown by a factor of 40!!

If want performance degradation < 10 percent

220 > 200 + 7,999,800 x p
20 > 7,999,800 x p

p < .0000025

< one page fault in every 400,000 memory accesses

Demand Paging Optimizations

- Swap space I/O faster than file system I/O even if on the same device
  - Swap allocated in larger chunks, less management needed than file system
- Copy entire process image to swap space at process load time
  - Then page in and out of swap space
  - Used in older BSD Unix
- Demand page in from program binary on disk, but discard rather than paging out when freeing frame
  - Used in Solaris and current BSD
  - Still need to write to swap space
    - Pages not associated with a file (like stack and heap) – **anonymous memory**
    - Pages modified in memory but not yet written back to the file system
- Mobile systems
  - Typically don't support swapping
  - Instead, demand page from file system and reclaim read-only pages (such as code)

Copy-on-Write

Copy-on-Write (COW) allows both parent and child processes to initially ***share*** the same pages in memory

- If either process modifies a shared page, only then is the page copied

COW allows more efficient process creation as only modified pages are copied

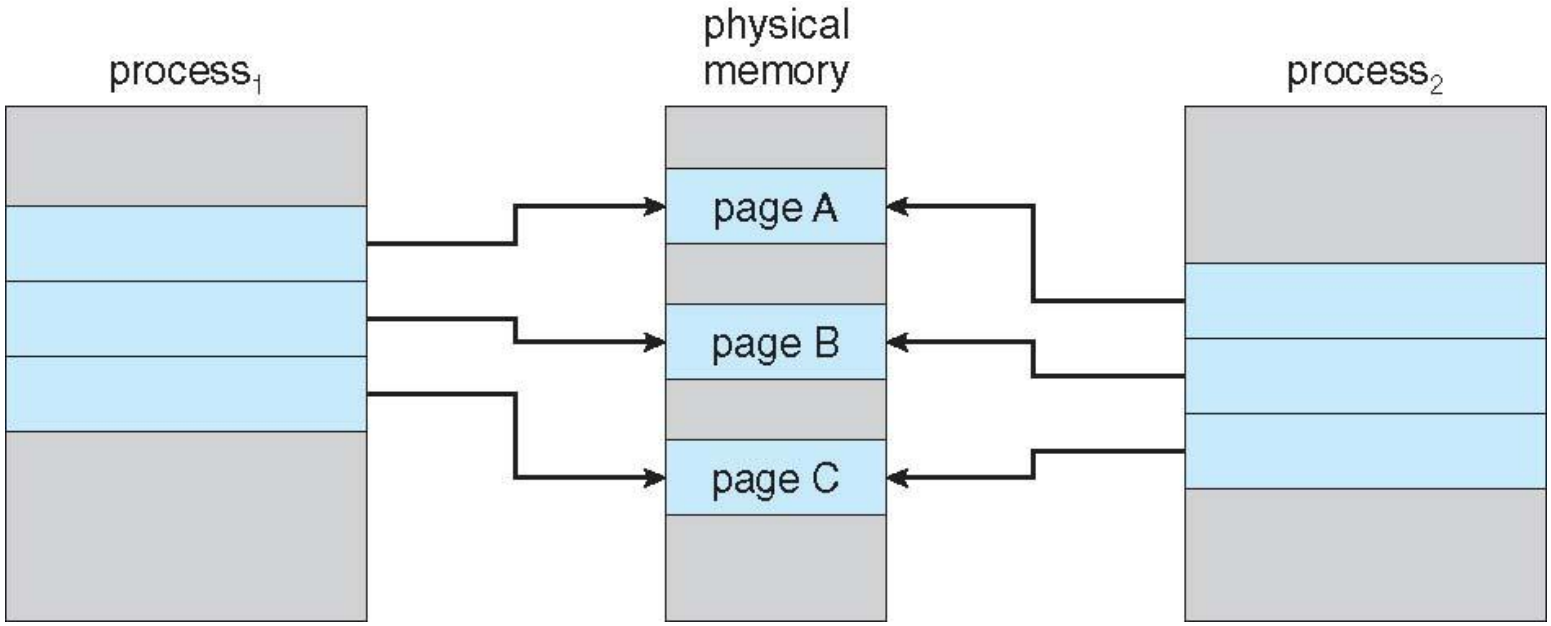
In general, free pages are allocated from a **pool** of **zero-fill-on-demand** pages

- Pool should always have free frames for fast demand page execution
  - Don't want to have to free a frame as well as other processing on page fault
- Why zero-out a page before allocating it?

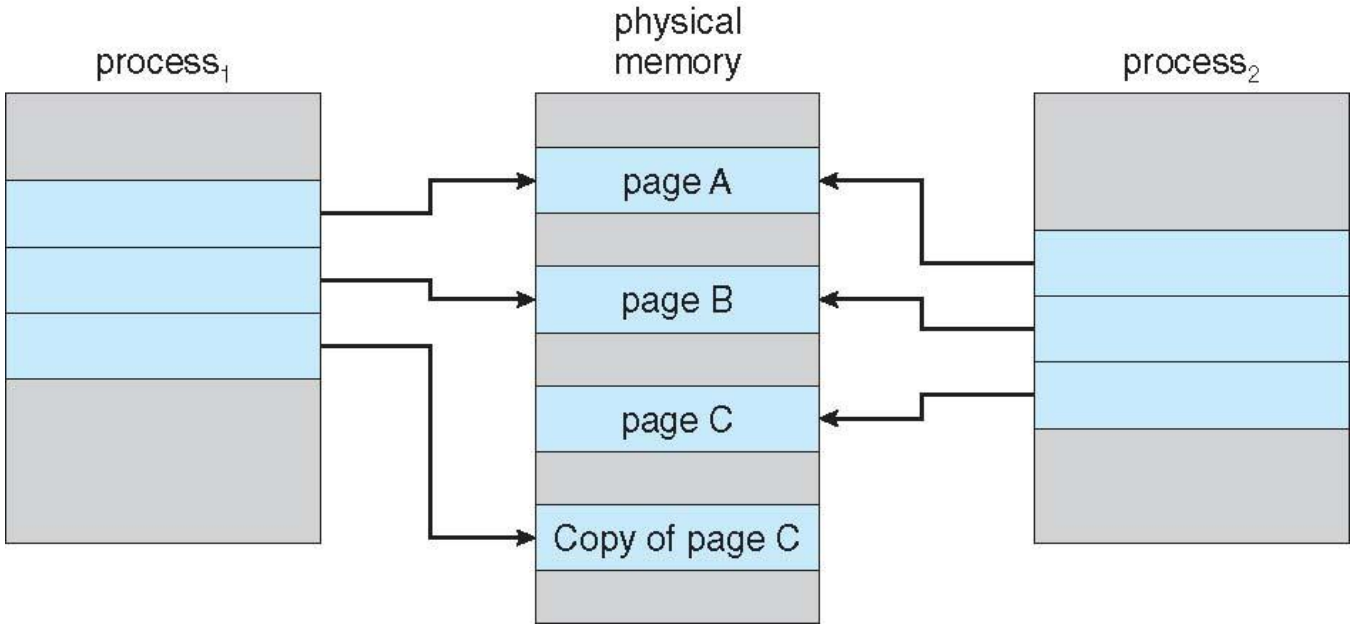
`vfork()` variation on `fork()` system call has parent suspend and child using copy-on-write address space of parent

- Designed to have child call `exec()`
- Very efficient

Before Process 1 Modifies Page C



After Process 1 Modifies Page C



What Happens if There is no Free Frame?

- Used up by process pages
- Also in demand from the kernel, I/O buffers, etc
- How much to allocate to each?
- Page replacement – find some page in memory, but not really in use, page it out
  - Algorithm – terminate? swap out? replace the page?
  - Performance – want an algorithm which will result in minimum number of page faults
- Same page may be brought into memory several times

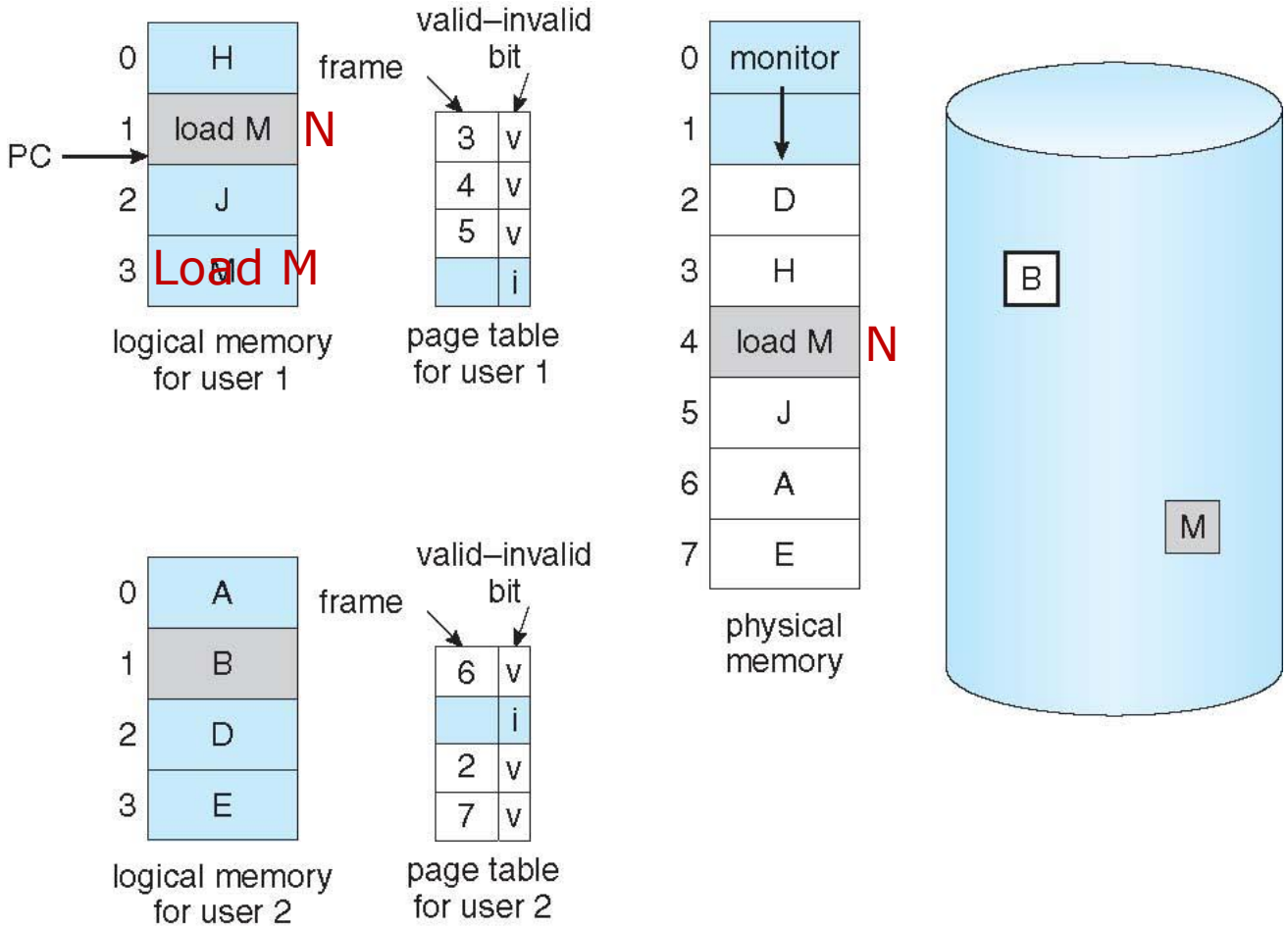
Page Replacement

Prevent **over-allocation** of memory by modifying page-fault service routine to include page replacement

Use **modify** (**dirty**) **bit** to reduce overhead of page transfers – only modified pages are written to disk

Page replacement completes separation between logical memory and physical memory – large virtual memory can be provided on a smaller physical memory

Need For Page Replacement



Basic Page Replacement

1. Find the location of the desired page on disk

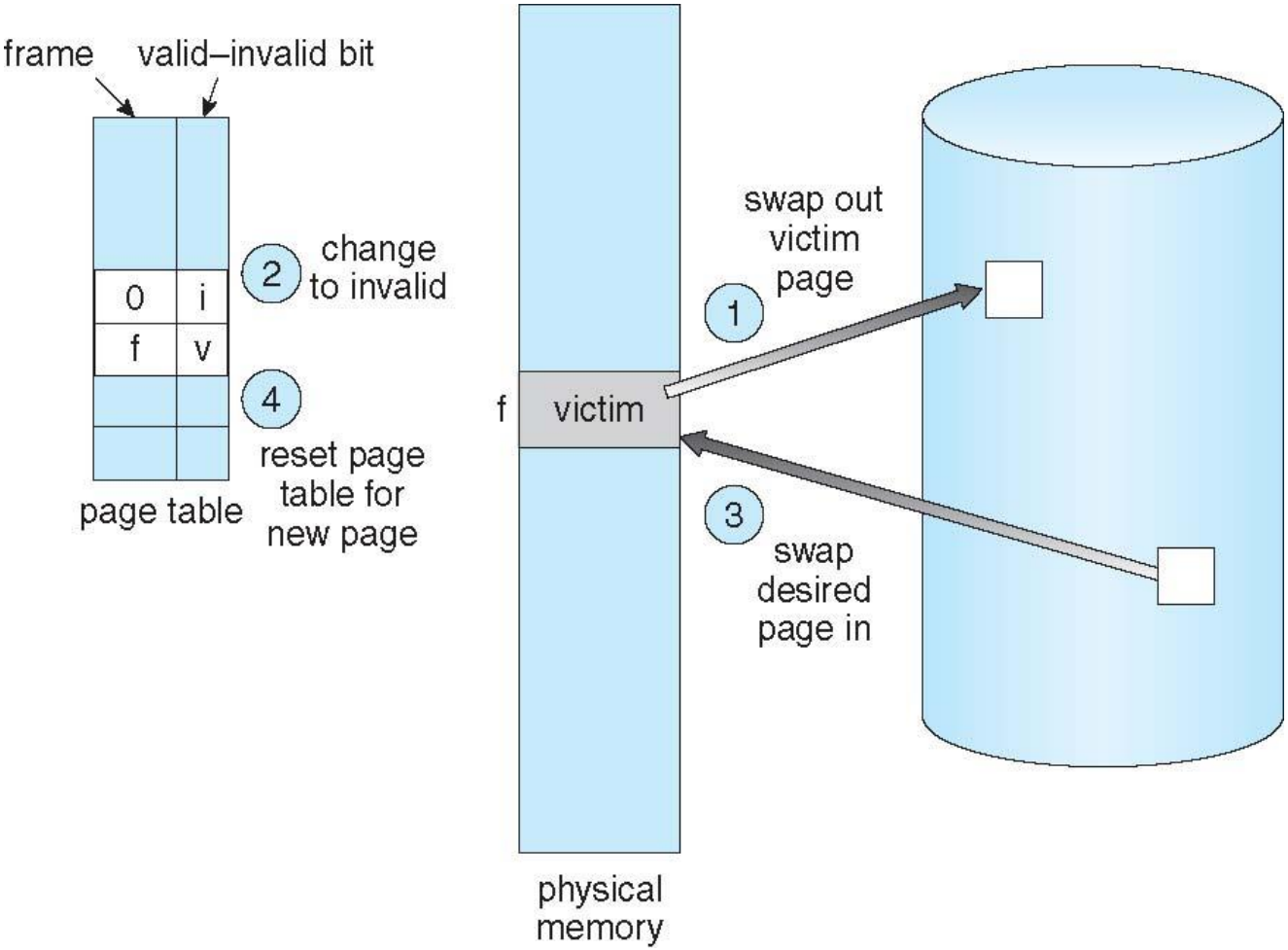
2. Find a free frame:
   - If there is a free frame, use it
   - If there is no free frame, use a page replacement algorithm to select a **victim frame**
     - Write victim frame to disk if dirty

3. Bring the desired page into the (newly) free frame; update the page and frame tables

4. Continue the process by restarting the instruction that caused the trap

Note now potentially 2 page transfers for page fault – increasing EAT

Page Replacement



Page and Frame Replacement Algorithms

Frame-allocation algorithm determines

- How many frames to give each process
- Which frames to replace

Page-replacement algorithm

- Want lowest page-fault rate on both first access and re-access

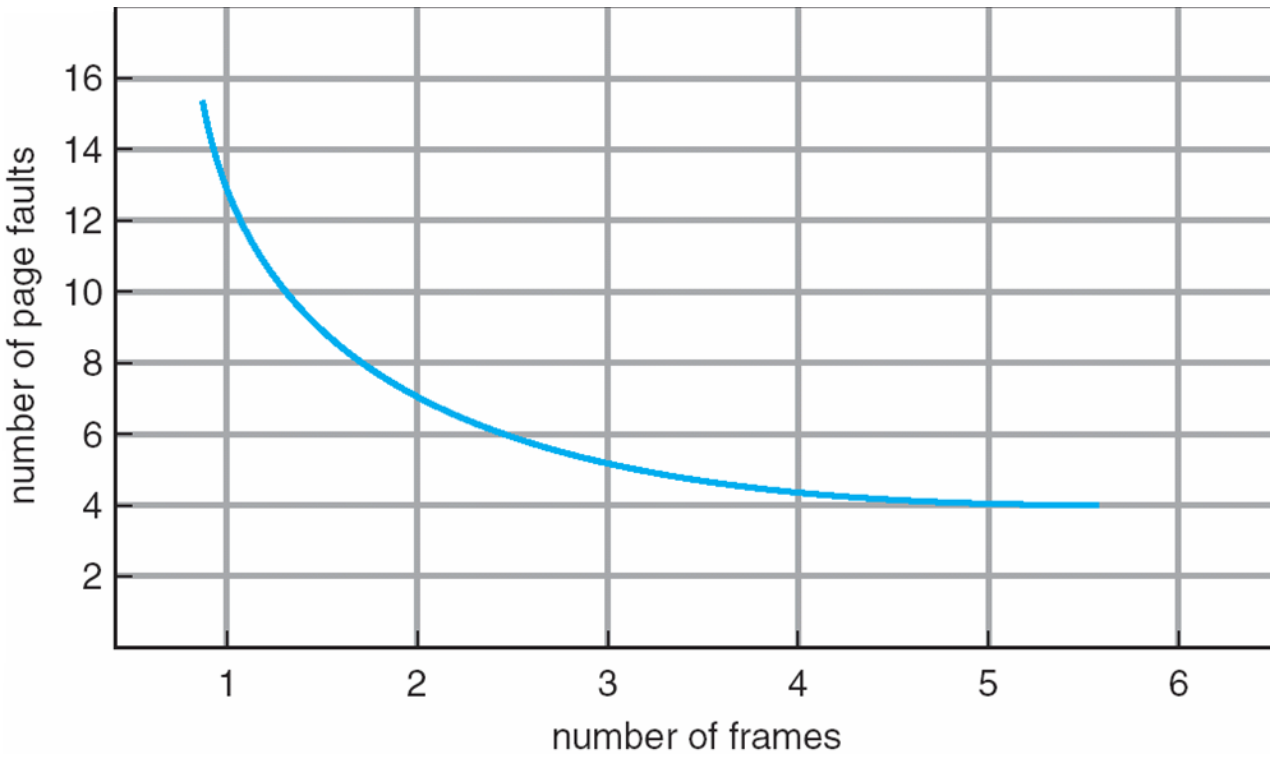
Evaluate algorithm by running it on a particular string of memory references (reference string) and computing the number of page faults on that string

- String is just page numbers, not full addresses
- Repeated access to the same page does not cause a page fault
- Results depend on number of frames available

In all our examples, the **reference string** of referenced page numbers is

7,0,1,2,0,3,0,4,2,3,0,3,0,3,2,1,2,0,1,7,0,1

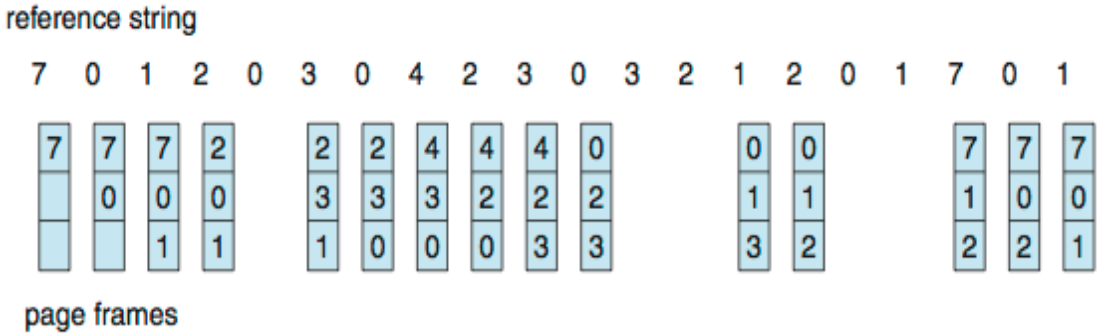
Graph of Page Faults Versus The Number of Frames



First-In-First-Out (FIFO) Algorithm

Reference string: **7,0,1,2,0,3,0,4,2,3,0,3,0,3,2,1,2,0,1,7,0,1**

3 frames (3 pages can be in memory at a time per process)



15 page faults

Can vary by reference string: consider 1,2,3,4,1,2,5,1,2,3,4,5

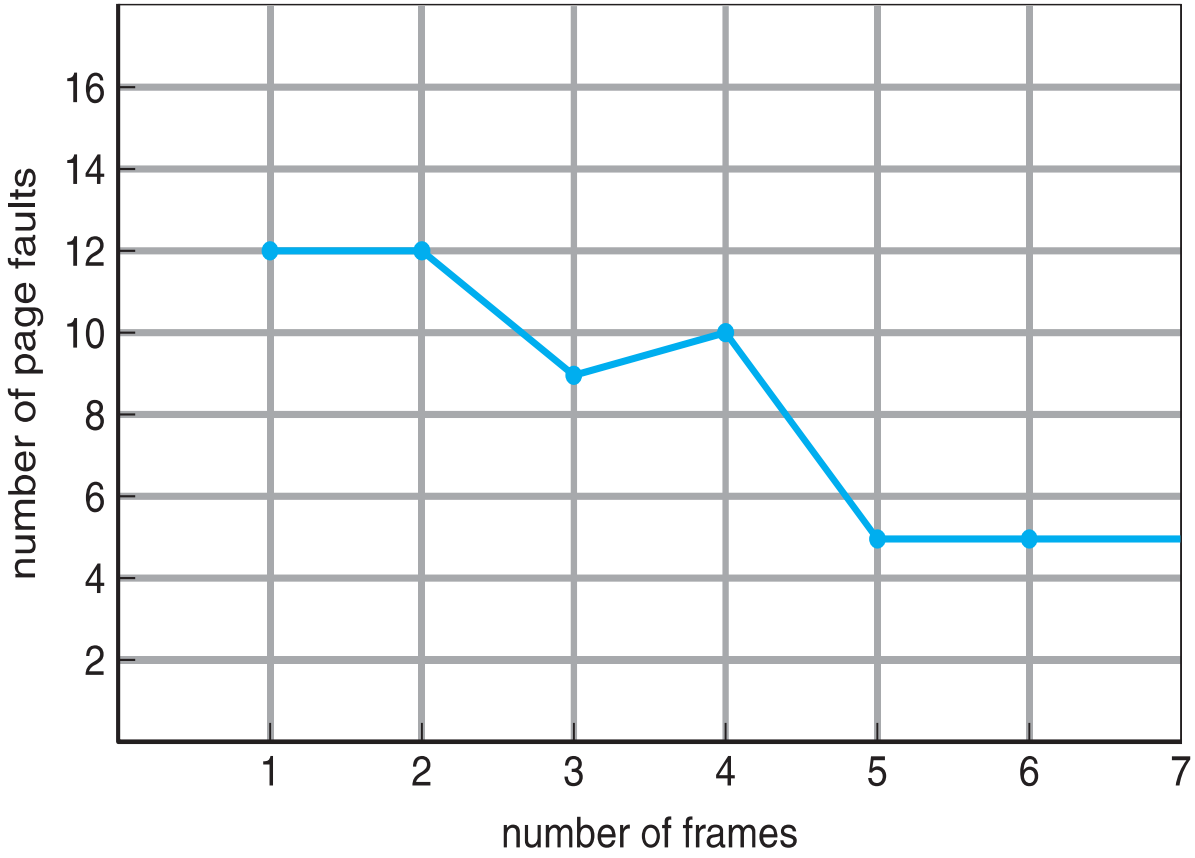
Adding more frames can cause more page faults!

- **Belady's Anomaly**

How to track ages of pages?

Just use a FIFO queue

FIFO Illustrating Belady's Anomaly



Optimal Algorithm

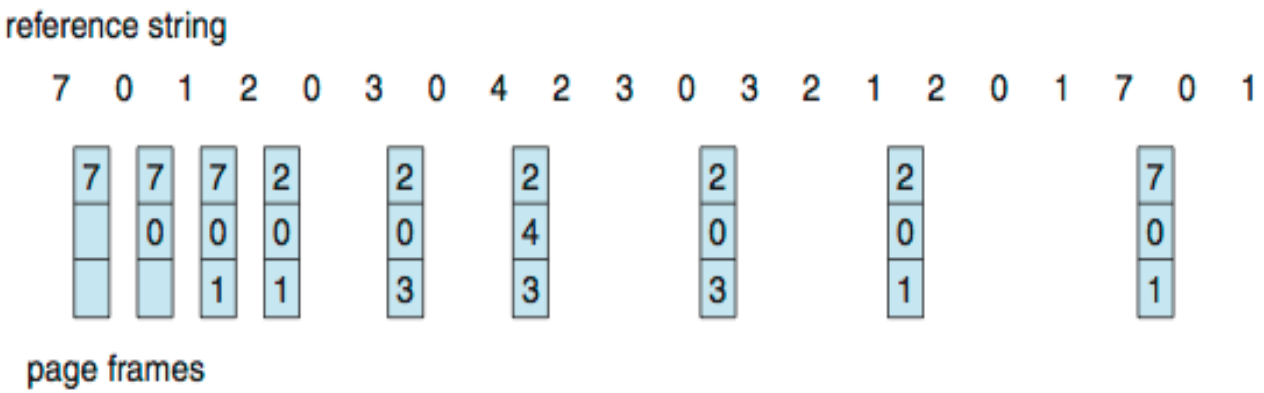
Replace page that will not be used for longest period of time

- 9 is optimal for the example

How do you know this?

- Can't read the future

Used for measuring how well your algorithm performs

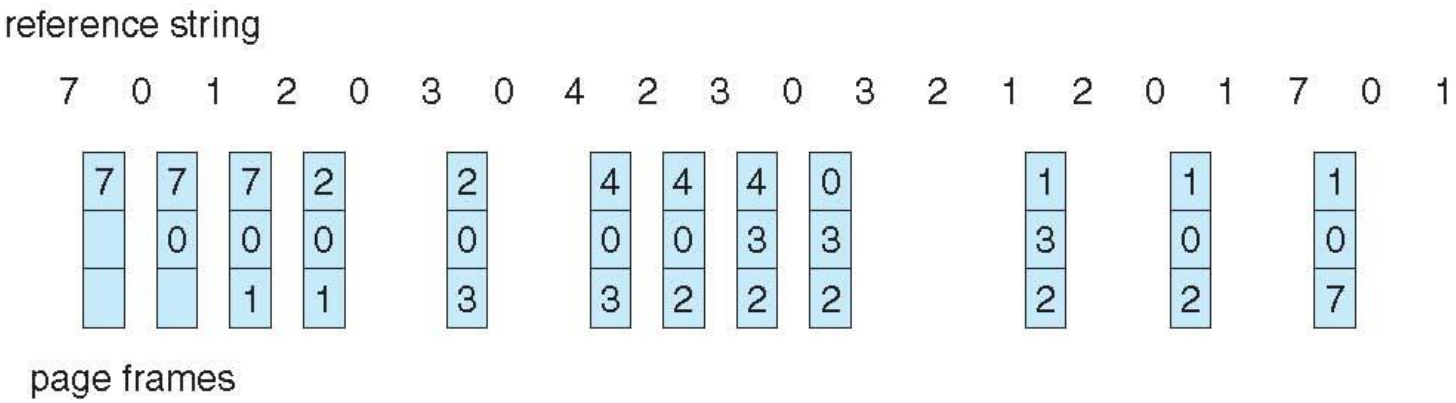


Least Recently Used (LRU) Algorithm

Use past knowledge rather than future

Replace page that has not been used in the most amount of time

Associate time of last use with each page



12 faults – better than FIFO but worse than OPT

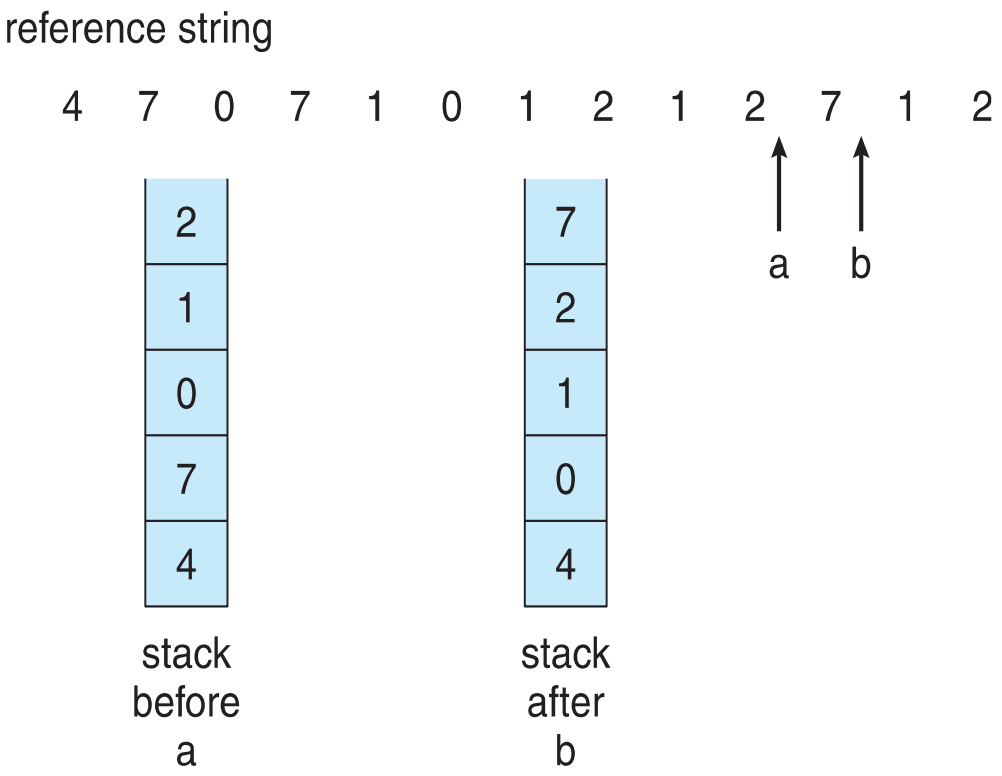
Generally good algorithm and frequently used

But how to implement?

LRU Algorithm (Cont.)

- Counter implementation
  - Every page entry has a counter; every time page is referenced through this entry, copy the clock into the counter
  - When a page needs to be changed, look at the counters to find smallest value
    - Search through table needed
- Stack implementation
  - Keep a stack of page numbers in a double link form:
  - Page referenced:
    - move it to the top
    - requires 6 pointers to be changed
  - But each update more expensive
  - No search for replacement
- LRU and OPT are cases of **stack algorithms** that don't have Belady's Anomaly

Use Of A Stack to Record Most Recent Page References



LRU Approximation Algorithms

LRU needs special hardware and still slow

Reference bit

With each page associate a bit, initially = 0

When page is referenced bit set to 1

Replace any with reference bit = 0 (if one exists)

- We do not know the order, however

Second-chance algorithm

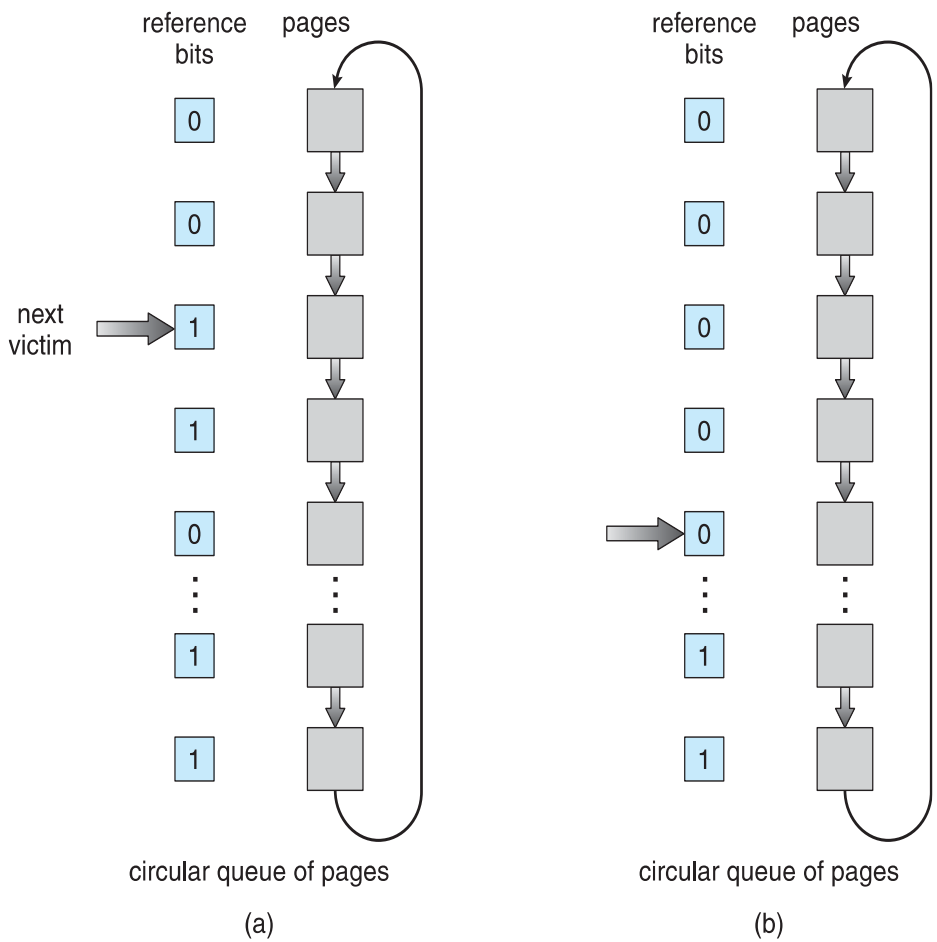
Generally FIFO, plus hardware-provided reference bit

Clock replacement

If page to be replaced has

- Reference bit = 0 -> replace it
- reference bit = 1 then:
  - set reference bit 0, leave page in memory
  - replace next page, subject to same rules

Second-Chance (clock) Page-Replacement Algorithm



Enhanced Second-Chance Algorithm

Improve algorithm by using reference bit and modify bit (if available) in concert

Take ordered pair (reference, modify)

1. (0, 0) neither recently used not modified – best page to replace
2. (0, 1) not recently used but modified – not quite as good, must write out before replacement
3. (1, 0) recently used but clean – probably will be used again soon
4. (1, 1) recently used and modified – probably will be used again soon and need to write out before replacement

When page replacement called for, use the clock scheme but use the four classes replace page in lowest non-empty class

Might need to search circular queue several times

Counting Algorithms

Keep a counter of the number of references that have been made to each page

Not common

Lease Frequently Used (**LFU**) **Algorithm**: replaces page with smallest count

Most Frequently Used (**MFU**) **Algorithm**: based on the argument that the page with the smallest count was probably just brought in and has yet to be used

Page-Buffering Algorithms

- Keep a pool of free frames, always
  - Then frame available when needed, not found at fault time
  - Read page into free frame and select victim to evict and add to free pool
  - When convenient, evict victim
- Possibly, keep list of modified pages
  - When backing store otherwise idle, write pages there and set to non-dirty
- Possibly, keep free frame contents intact and note what is in them
  - If referenced again before reused, no need to load contents again from disk
  - Generally useful to reduce penalty if wrong victim frame selected

Applications and Page Replacement

All of these algorithms have OS guessing about future page access

Some applications have better knowledge – i.e. databases

Memory intensive applications can cause double buffering

- OS keeps copy of page in memory as I/O buffer
- Application keeps page in memory for its own work

Operating system can given direct access to the disk, getting out of the way of the applications

- **Raw disk** mode

Bypasses buffering, locking, etc

Allocation of Frames

Each process needs ***minimum*** number of frames

Example: IBM 370 – 6 pages to handle SS MOVE instruction:

- instruction is 6 bytes, might span 2 pages
- 2 pages to handle *from*
- 2 pages to handle *to*

Maximum of course is total frames in the system

Two major allocation schemes

- fixed allocation
- priority allocation

Many variations

Fixed Allocation

Equal allocation – For example, if there are 100 frames (after allocating frames for the OS) and 5 processes, give each process 20 frames

Keep some as free frame buffer pool

Proportional allocation – Allocate according to the size of process

Dynamic as degree of multiprogramming, process sizes change

- s_i = size of process p_i
- $S = \sum s_i$
- m = total number of frames
- a_i = allocation for $p_i = \frac{s_i}{S} \times m$

$m = 64$

$s_1 = 10$

$s_2 = 127$

$a_1 = \frac{10}{137} \times 62 \approx 4$

$a_2 = \frac{127}{137} \times 62 \approx 57$

Priority Allocation

Use a proportional allocation scheme using priorities rather than size

If process P_i generates a page fault,

- select for replacement one of its frames
- select for replacement a frame from a process with lower priority number

Global vs. Local Allocation

Global replacement – process selects a replacement frame from the set of all frames; one process can take a frame from another

- But then process execution time can vary greatly
- But greater throughput so more common

Local replacement – each process selects from only its own set of allocated frames

- More consistent per-process performance
- But possibly underutilized memory

Non-Uniform Memory Access

So far all memory accessed equally

Many systems are **NUMA** – speed of access to memory varies

Consider system boards containing CPUs and memory, interconnected over a system bus

Optimal performance comes from allocating memory "close to" the CPU on which the thread is scheduled

And modifying the scheduler to schedule the thread on the same system board when possible

Solved by Solaris by creating **lgroups**

- Structure to track CPU / Memory low latency groups
- Used my schedule and pager
- When possible schedule all threads of a process and allocate all memory for that process within the lgroup

Thrashing

If a process does not have "enough" pages, the page-fault rate is very high

Page fault to get page

Replace existing frame

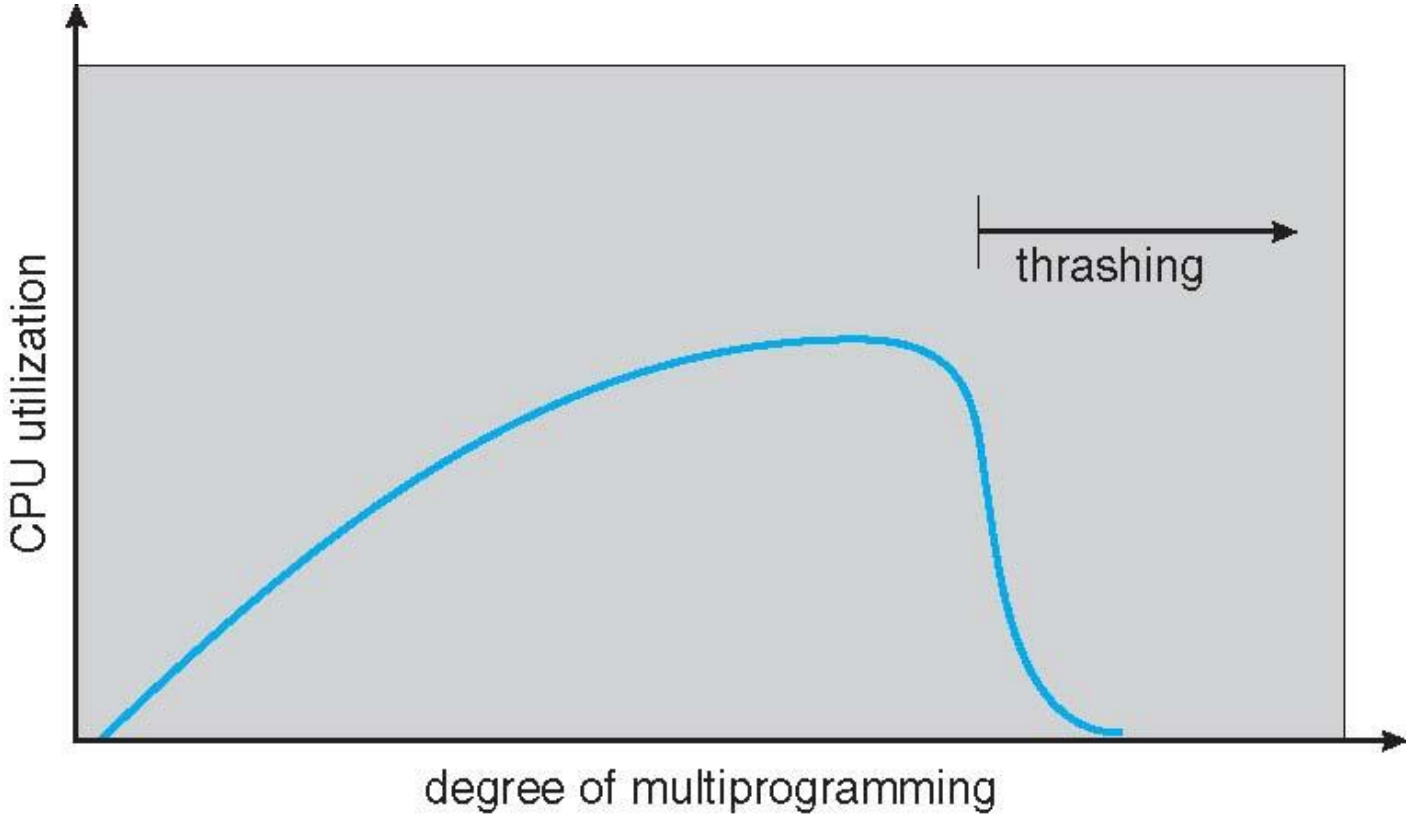
But quickly need replaced frame back

This leads to:

- Low CPU utilization
- Operating system thinking that it needs to increase the degree of multiprogramming
- Another process added to the system

Thrashing ≡ a process is busy swapping pages in and out

Thrashing (Cont.)



Demand Paging and Thrashing

Why does demand paging work?

Locality model

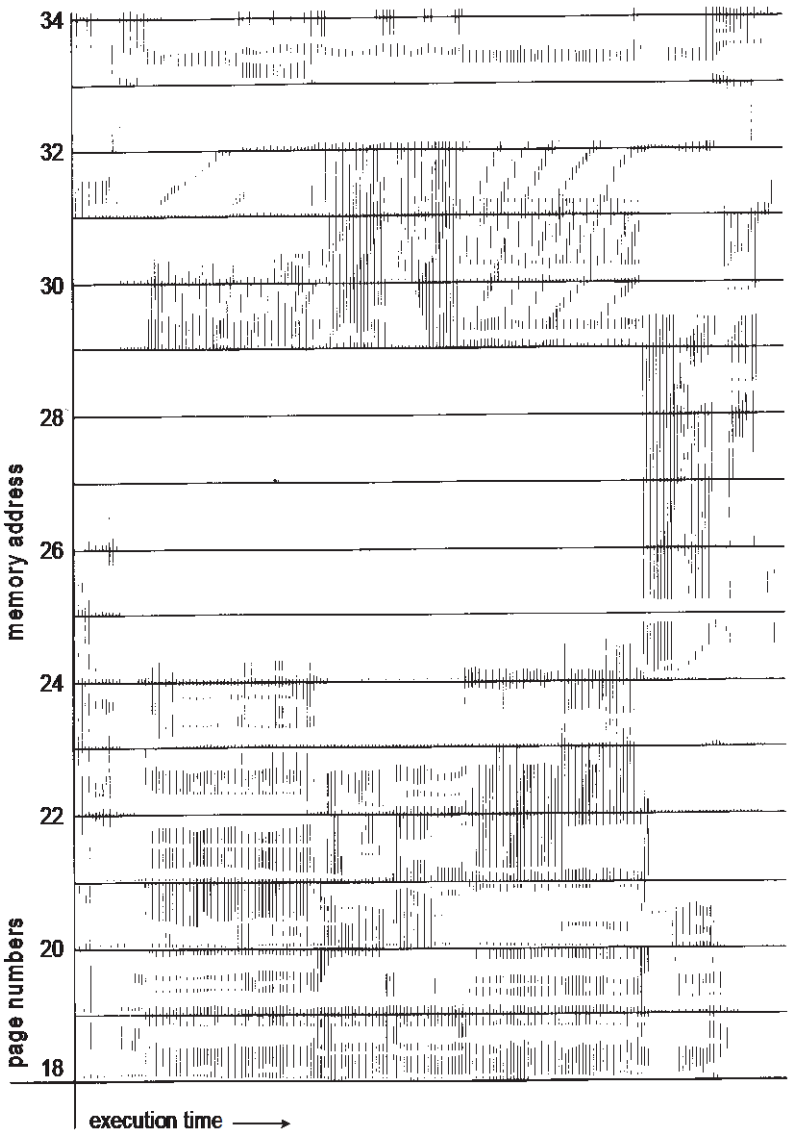
- Process migrates from one locality to another
- Localities may overlap

Why does thrashing occur?

Σ size of locality > total memory size

- Limit effects by using local or priority page replacement

Locality In A Memory-Reference Pattern



Working-Set Model

$\Delta \equiv$ working-set window $\equiv$ a fixed number of page references
Example: 10,000 instructions

WSS_i (working set of Process P_i) =
total number of pages referenced in the most recent Δ (varies in time)

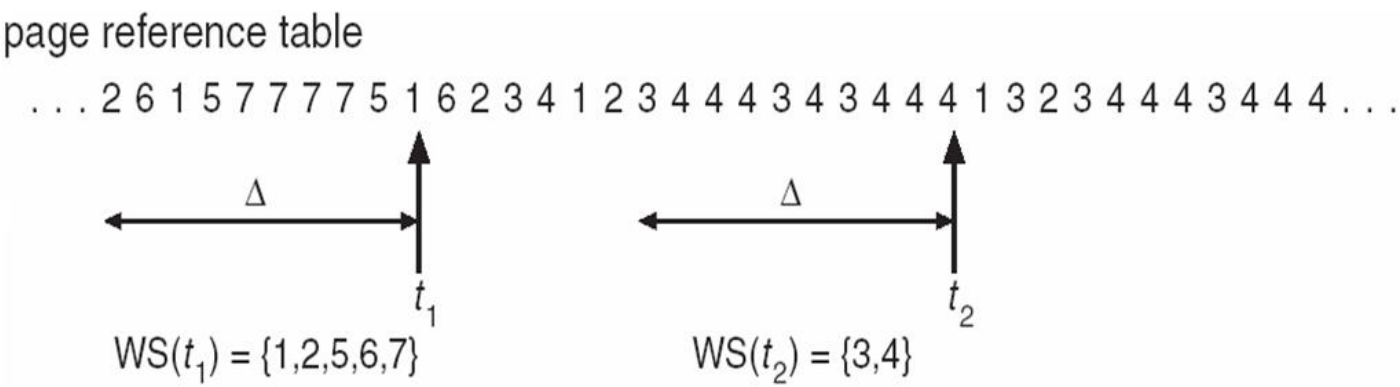
- if Δ too small will not encompass entire locality
- if Δ too large will encompass several localities
- if $\Delta = \infty \Rightarrow$ will encompass entire program

$D = \Sigma\ WSS_i \equiv$ total demand frames

- Approximation of locality

if $D > m \Rightarrow$ Thrashing

Policy if D > m, then suspend or swap out one of the processes



Keeping Track of the Working Set

Approximate with interval timer + a reference bit

Example: Δ = 10,000

- Timer interrupts after every 5000 time units
- Keep in memory 2 bits for each page
- Whenever a timer interrupts copy and sets the values of all reference bits to 0
- If one of the bits in memory = 1 $\Rightarrow$ page in working set

Why is this not completely accurate?

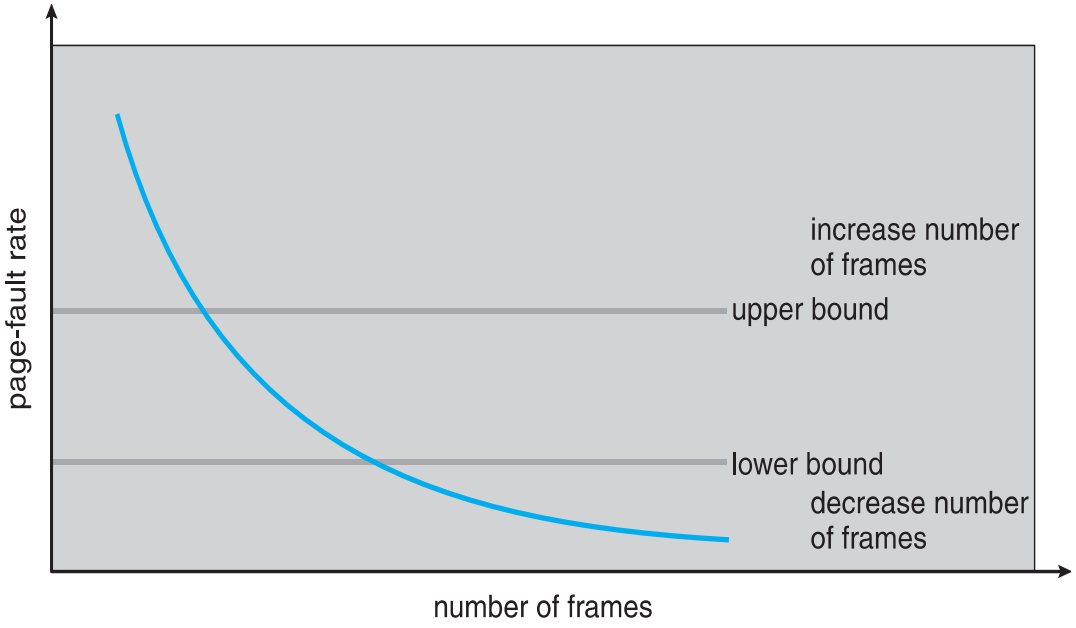
Improvement = 10 bits and interrupt every 1000 time units

Page-Fault Frequency

More direct approach than WSS

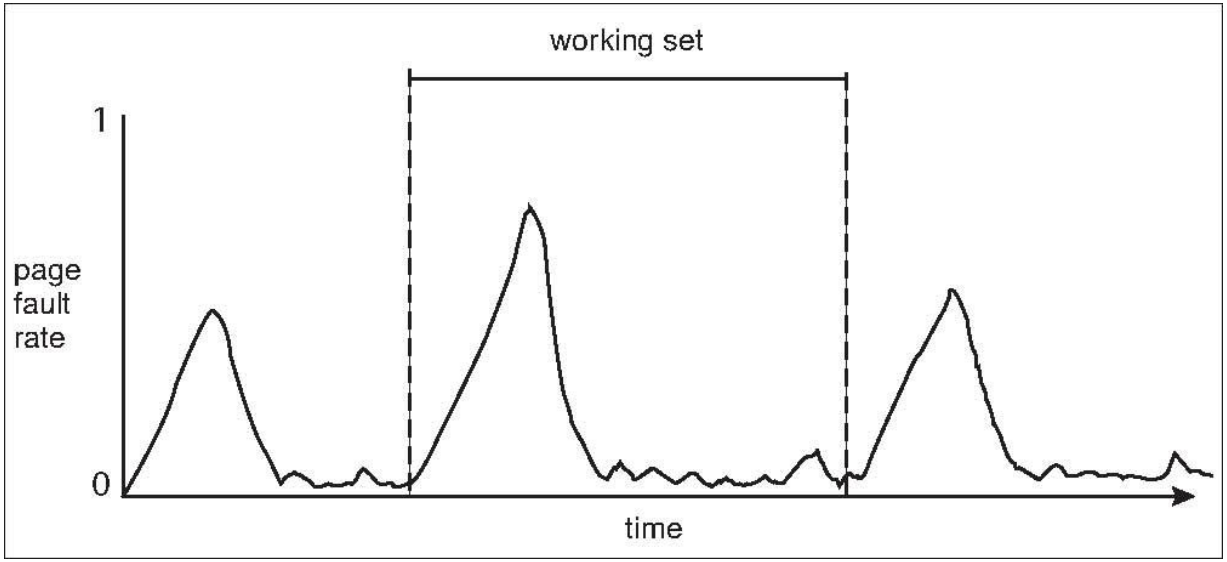
Establish "acceptable" **page-fault frequency** (**PFF**) rate and use local replacement policy

- If actual rate too low, process loses frame
- If actual rate too high, process gains frame



Working Sets and Page Fault Rates

- Direct relationship between working set of a process and its page-fault rate
- Working set changes over time
- Peaks and valleys over time



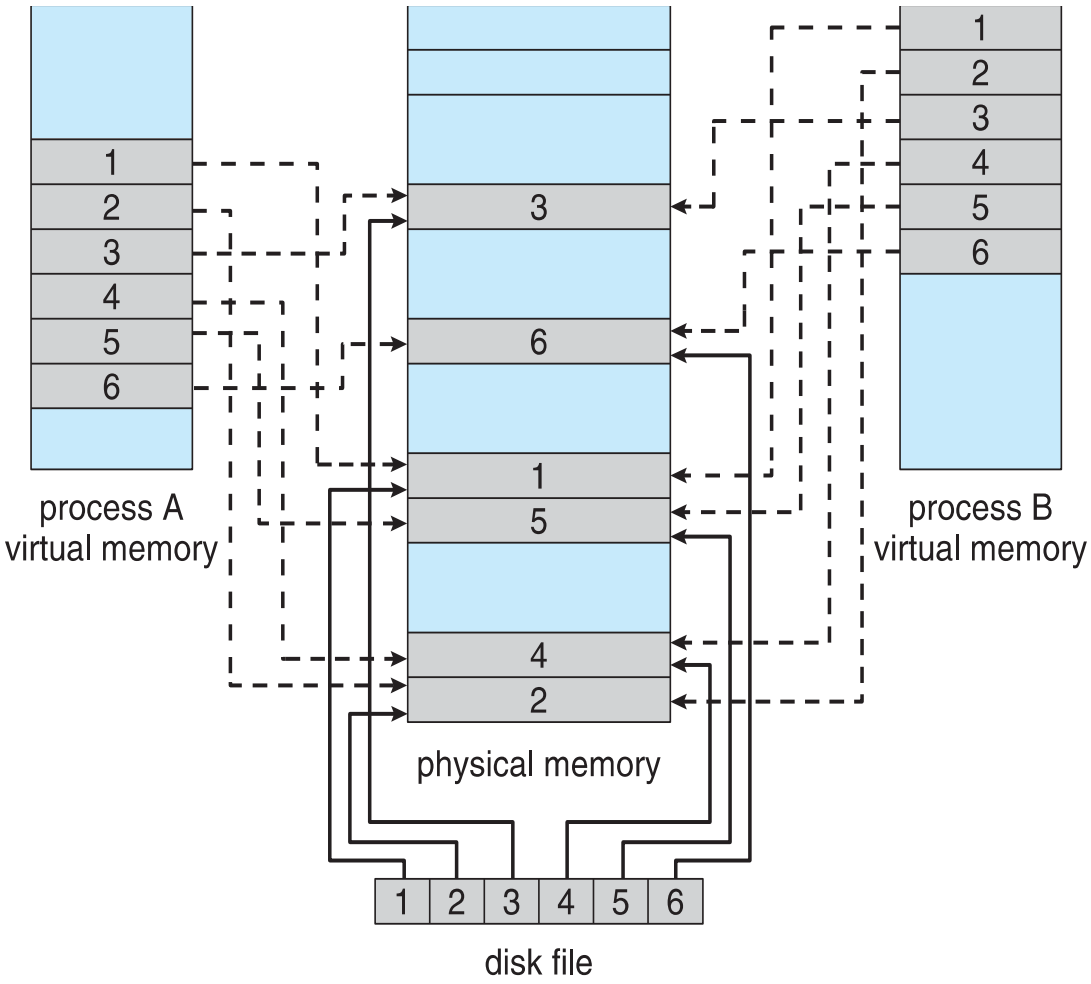
Memory-Mapped Files

- Memory-mapped file I/O allows file I/O to be treated as routine memory access by **mapping** a disk block to a page in memory
- A file is initially read using demand paging
  - A page-sized portion of the file is read from the file system into a physical page
  - Subsequent reads/writes to/from the file are treated as ordinary memory accesses
- Simplifies and speeds file access by driving file I/O through memory rather than `read()` and `write()` system calls
- Also allows several processes to map the same file allowing the pages in memory to be shared
- But when does written data make it to disk?
  - Periodically and / or at file `close()` time
  - For example, when the pager scans for dirty pages

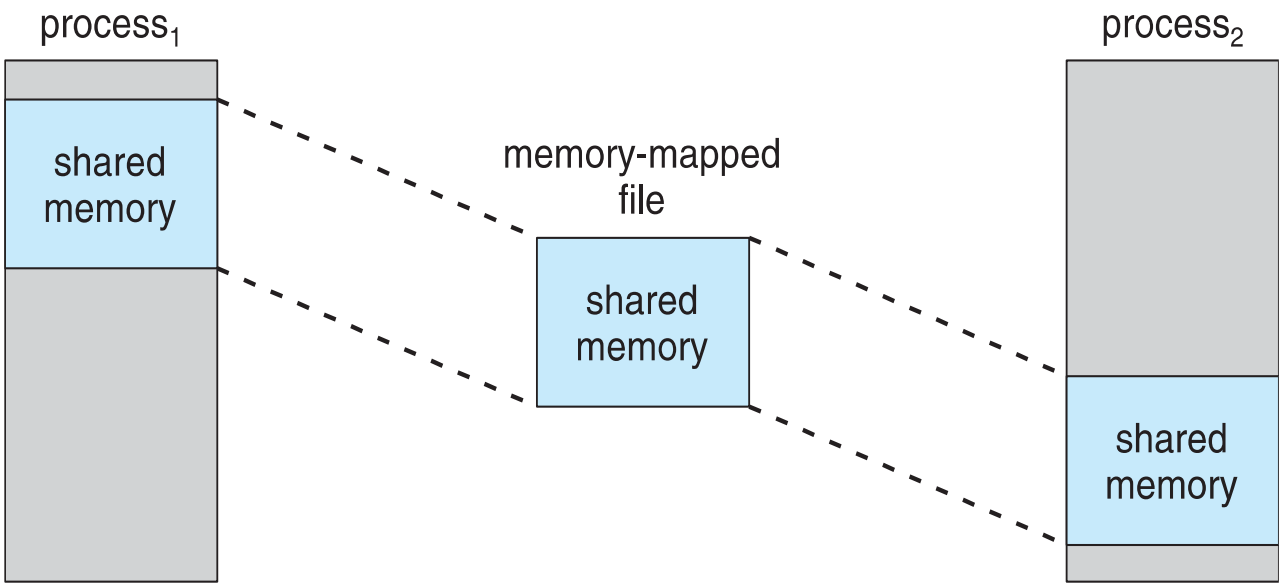
Memory-Mapped File Technique for all I/O

- Some OSes uses memory mapped files for standard I/O
- Process can explicitly request memory mapping a file via `mmap()` system call
  - Now file mapped into process address space
- For standard I/O (`open()`, `read()`, `write()`, `close()`), mmap anyway
  - But map file into kernel address space
  - Process still does read() and write()
    - Copies data to and from kernel space and user space
  - Uses efficient memory management subsystem
    - Avoids needing separate subsystem
- COW can be used for read/write non-shared pages
- Memory mapped files can be used for shared memory (although again via separate system calls)

Memory Mapped Files



Shared Memory via Memory-Mapped I/O



Shared Memory in Windows API

- First create a **file mapping** for file to be mapped
  - Then establish a view of the mapped file in process's virtual address space
- Consider producer / consumer
  - Producer create shared-memory object using memory mapping features
  - Open file via `CreateFile()`, returning a `HANDLE`
  - Create mapping via `CreateFileMapping()` creating a **named shared-memory object**
  - Create view via `MapViewOfFile()`
- Sample code in Textbook

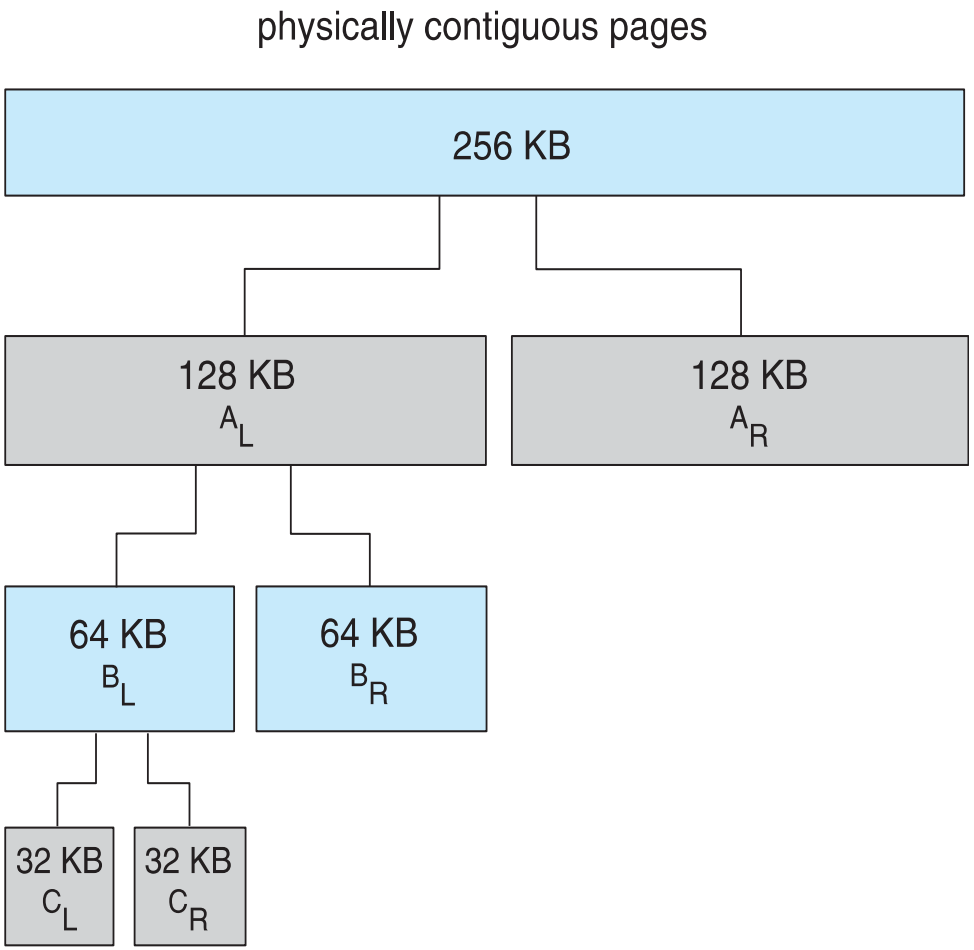
Allocating Kernel Memory

- Treated differently from user memory
- Often allocated from a free-memory pool
  - Kernel requests memory for structures of varying sizes
  - Some kernel memory needs to be contiguous
    - I.e. for device I/O

Buddy System

- Allocates memory from fixed-size segment consisting of physically-contiguous pages
- Memory allocated using **power-of-2 allocator**
  - Satisfies requests in units sized as power of 2
  - Request rounded up to next highest power of 2
  - When smaller allocation needed than is available, current chunk split into two buddies of next-lower power of 2
    - Continue until appropriate sized chunk available
- For example, assume 256KB chunk available, kernel requests 21KB
  - Split into A_L and A_R of 128KB each
    - One further divided into B_L and B_R of 64KB
      - One further into C_L and C_R of 32KB each – one used to satisfy request
- Advantage – quickly **coalesce** unused chunks into larger chunk
- Disadvantage - fragmentation

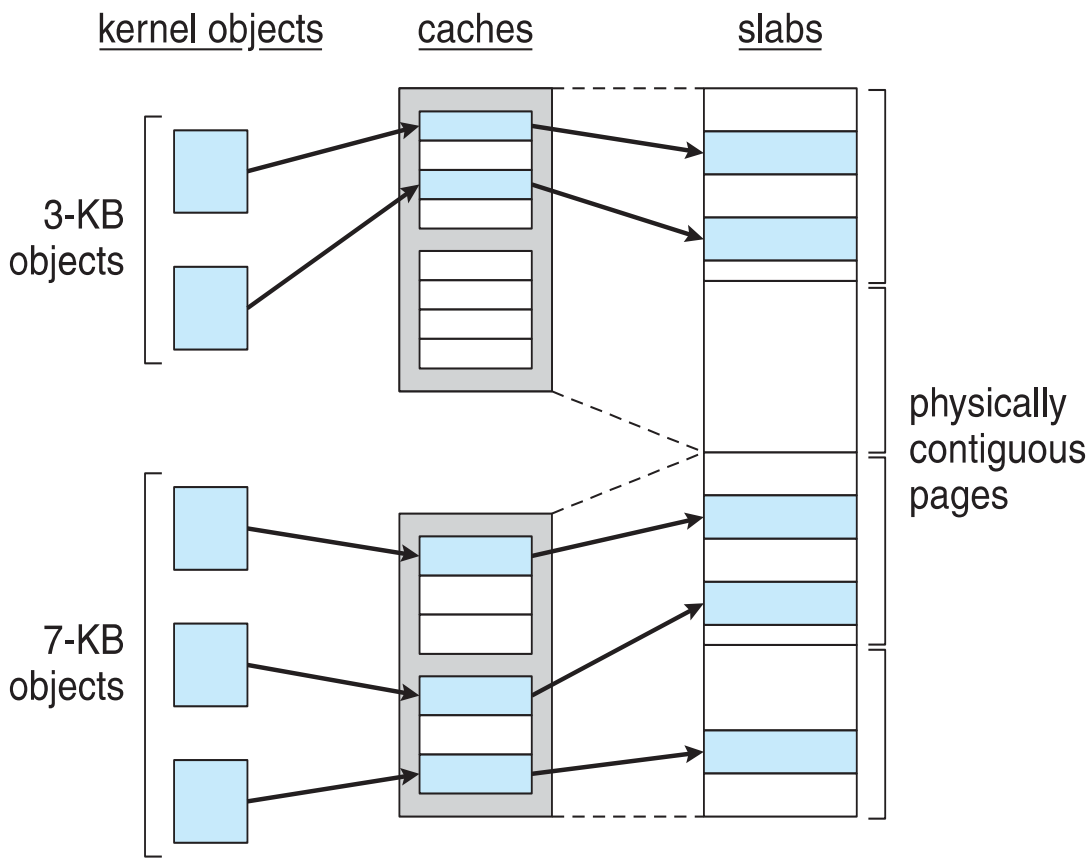
Buddy System Allocator



Slab Allocator

- Alternate strategy
- **Slab** is one or more physically contiguous pages
- **Cache** consists of one or more slabs
- Single cache for each unique kernel data structure
  - Each cache filled with **objects** – instantiations of the data structure
- When cache created, filled with objects marked as **`free`**
- When structures stored, objects marked as **`used`**
- If slab is full of used objects, next object allocated from empty slab
  - If no empty slabs, new slab allocated
- Benefits include no fragmentation, fast memory request satisfaction

Slab Allocation



Slab Allocator in Linux

For example process descriptor is of type `struct task_struct`

Approx 1.7KB of memory

New task -> allocate new struct from cache

- Will use existing free `struct task_struct`

Slab can be in three possible states

1. Full – all used
2. Empty – all free
3. Partial – mix of free and used

Upon request, slab allocator

1. Uses free struct in partial slab
2. If none, takes one from empty slab
3. If no empty slab, create new empty

Slab Allocator in Linux (Cont.)

Slab started in Solaris, now wide-spread for both kernel mode and user memory in various OSes

Linux 2.2 had SLAB, now has both SLOB and SLUB allocators

SLOB for systems with limited memory

- Simple List of Blocks – maintains 3 list objects for small, medium, large objects

SLUB is performance-optimized SLAB removes per-CPU queues, metadata stored in page structure

Other Considerations -- Prepaging

Prepaging

- To reduce the large number of page faults that occurs at process startup
- Prepage all or some of the pages a process will need, before they are referenced
- But if prepaged pages are unused, I/O and memory was wasted
- Assume *s* pages are prepaged and *α* of the pages is used
  - Is cost of ***s * α*** save pages faults > or < than the cost of prepaging
    s * (1- α) unnecessary pages?
  - ***α*** near zero ⇒ prepaging loses

Other Issues – Page Size

- Sometimes OS designers have a choice
  - Especially if running on custom-built CPU
- Page size selection must take into consideration:
  - Fragmentation
  - Page table size
  - **Resolution**
  - I/O overhead
  - Number of page faults
  - Locality
  - TLB size and effectiveness
- Always power of 2, usually in the range 2^{12} (4,096 bytes) to 2^{22} (4,194,304 bytes)
- On average, growing over time

Other Issues – TLB Reach

TLB Reach - The amount of memory accessible from the TLB

TLB Reach = (TLB Size) X (Page Size)

Ideally, the working set of each process is stored in the TLB

- Otherwise there is a high degree of page faults

Increase the Page Size

- This may lead to an increase in fragmentation as not all applications require a large page size

Provide Multiple Page Sizes

- This allows applications that require larger page sizes the opportunity to use them without an increase in fragmentation

Other Issues – Program Structure

Program structure

```
int[128,128] data;
```

Each row is stored in one page

Program 1

```
for (j = 0; j <128; j++)
    for (i = 0; i < 128; i++)
        data[i,j] = 0;
```

128 x 128 = 16,384 page faults

Program 2

```
for (i = 0; i < 128; i++)
    for (j = 0; j < 128; j++)
        data[i,j] = 0;
```

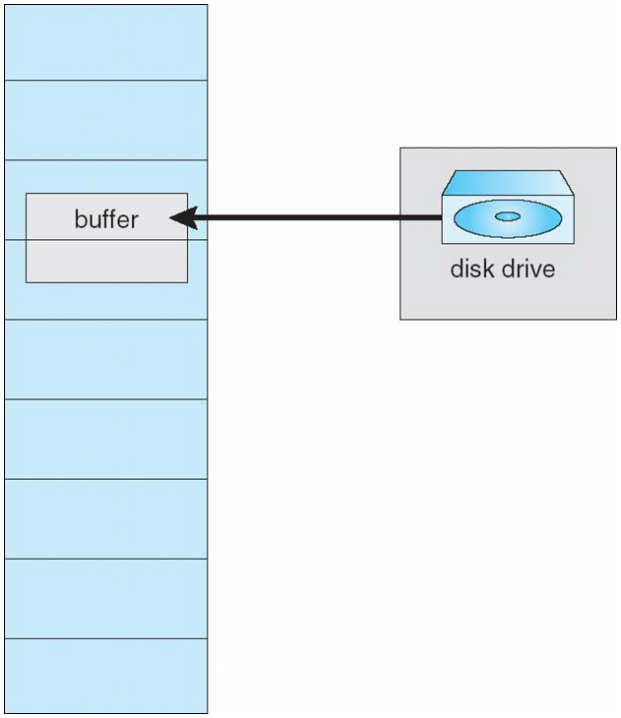
128 page faults

Other Issues – I/O interlock

I/O Interlock – Pages must sometimes be locked into memory

Consider I/O - Pages that are used for copying a file from a device must be locked from being selected for eviction by a page replacement algorithm

Pinning of pages to lock into memory



Operating System Examples

Windows

Solaris

Windows

- Uses demand paging with **clustering**. Clustering brings in pages surrounding the faulting page
- Processes are assigned **working set minimum** and **working set maximum**
- Working set minimum is the minimum number of pages the process is guaranteed to have in memory
- A process may be assigned as many pages up to its working set maximum
- When the amount of free memory in the system falls below a threshold, **automatic working set trimming** is performed to restore the amount of free memory
- Working set trimming removes pages from processes that have pages in excess of their working set minimum

Solaris

Maintains a list of free pages to assign faulting processes

`Lotsfree` – threshold parameter (amount of free memory) to begin paging

`Desfree` – threshold parameter to increasing paging

`Minfree` – threshold parameter to being swapping

Paging is performed by **`pageout`** process

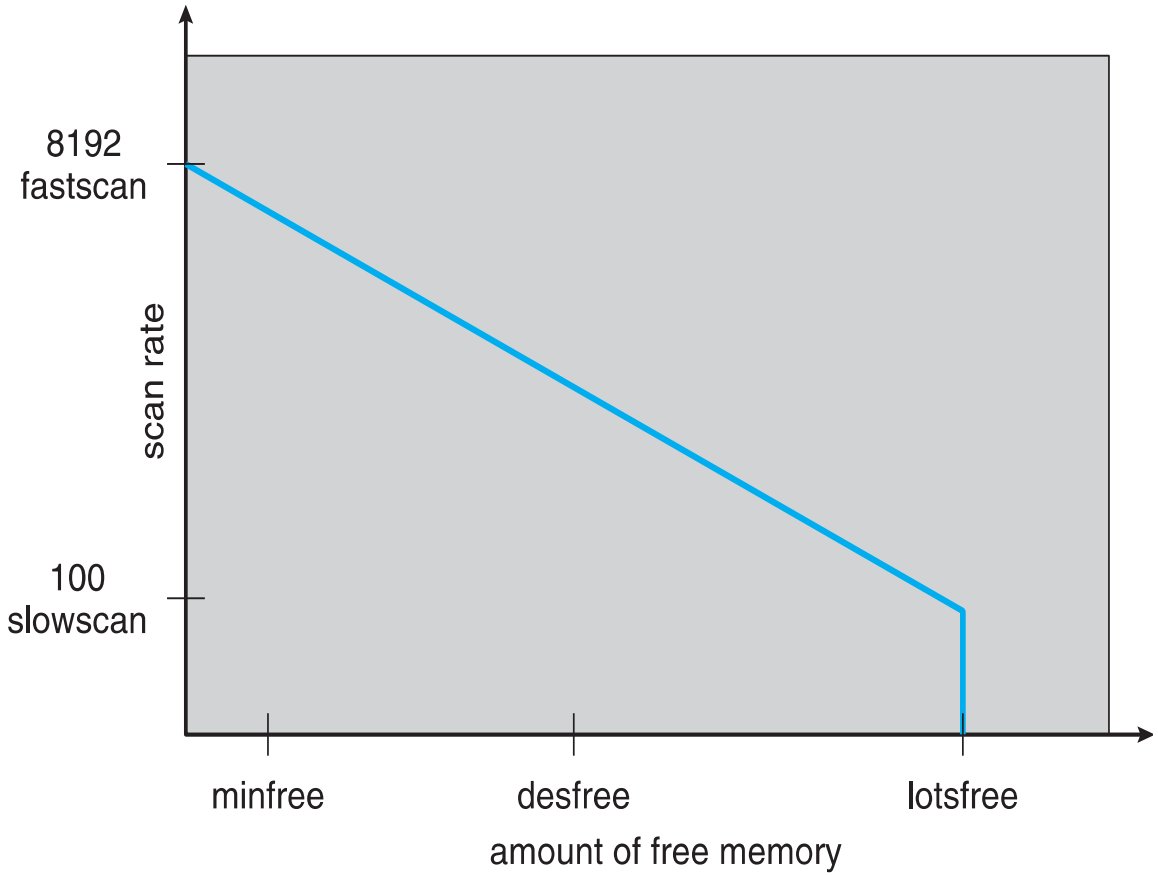
`Pageout` scans pages using modified clock algorithm

`Scanrate` is the rate at which pages are scanned. This ranges from **`slowscan`** to **`fastscan`**

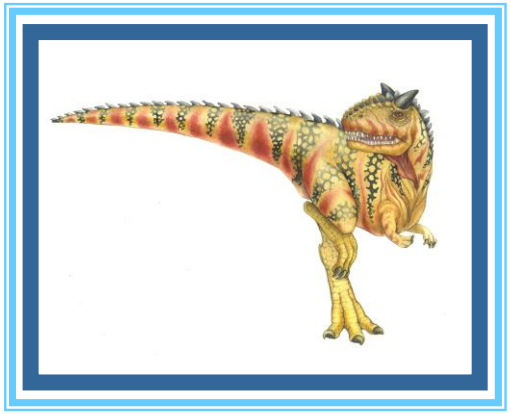
`Pageout` is called more frequently depending upon the amount of free memory available

Priority paging gives priority to process code pages

Solaris 2 Page Scanner



End of Chapter 10



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